

Looking Monitorable Without Being Faithful: Math-CoT Fine-Tuning Degrades Cue-Faithfulness Independent of Teacher Strength

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Abstract

We test whether fine-tuning a reasoning model on math CoT preserves chain-of-thought monitorability, using the cue-based faithfulness paradigm (Meek et al. 2025; Turpin et al. 2023) on a DeepSeek-R1-distilled student. Our central finding is a **dissociation between behavior and verbalization**: after weak-to-strong reasoning fine-tuning (W2SR; Yuan et al. 2025), a planted cue’s *behavioral* pull on the answer is *not reduced* (flips land on the cue target at $\sim 2\times$ chance, paired influence $\Delta = +0.16$), while the CoT’s *acknowledgment* collapses $25\% \rightarrow 3\%$ (34/1 one-directional discordant, $p \approx 2 \times 10^{-9}$) at improved GPQA accuracy ($0.28 \rightarrow 0.42$). Lobo et al. (2024) already showed fine-tuning lowers faithfulness on an answer-level metric; our contribution is that the cue’s causal grip is held fixed while verbalization disappears. The effect is **teacher-strength-invariant**: a same-strength self-distillation control (teacher == student) collapses acknowledgment indistinguishably from W2SR weak ($p = 1$, $n = 79$), so weak-to-strong asymmetry is incidental and the cause is math-CoT SFT in general. The raw drop is partly compression (CoT shortens $\sim 14\times$, matched-length residual marginal across all samples), but **restricted to the safety-relevant subset where the cue actually moved the answer** (influenced = 1), **W2SR independently suppresses cue acknowledgment at matched mid-length** (baseline 70% vs. W2SR 21%, OR ≈ 8.75 , $p = 0.003$). The effect is robust to an out-of-family judge ($\kappa \in [0.65, 0.68]$, paired drop preserved) and replicates on MMLU at $\sim 3\times$ compression with a -17pp drop — a within-paper dose-response. **A capability-only certification can pass a model whose CoT has gone behaviorally opaque while remaining superficially legible.**

1 Introduction

Chain-of-thought “monitorability” is the hope that reading a model’s reasoning reveals *why* it answered, so an overseer can catch undesired influences. Weak-to-Strong Reasoning (W2SR) [1] reports that fine-tuning a strong student on a *weaker* model’s CoT *raises accuracy*. Faithfulness work [3, 2, 4] shows models often do not verbalize the cues that actually move their answers.

Critique. W2SR certifies the student on accuracy alone and is silent on whether the distilled CoT stays monitorable. If distillation changed *how* a model reasons on the page without changing *what it reveals*, an accuracy check would miss it.

Contributions. The phenomena we exploit are largely known — Lobo et al. [5] showed fine-tuning hurts CoT faithfulness on an answer-level metric; Zaman & Srivastava [7] argued via causal mediation that the dissociation between behavior and CoT verbalization can exist statically. Our contributions are methodological and target-specific. **(1) Controllable induction of the dissociation via a specific training method.** We separate behavior (paired influence and flip-to-cue rate, $\sim 2\times$ chance) from verbalization (judge acknowledgment) and show that math-CoT SFT *controllably* moves an R1-distill student from monitorable (ack 25%, behavior preserved) to monitorable-looking-but-not (ack 3%, behavior still preserved). Beyond observing the dissociation, the contribution is that a marketed training method induces it on demand with behavior held fixed (§5). **(2) Same-strength self-distillation negative control.** A self-distillation arm (teacher == student, same R1-7B traces) reproduces the collapse indistinguishably from W2SR weak (paired $p = 1$, $n = 79$), with pairwise McNemar $p \geq 0.09$ between all trained arms. To our knowledge no prior work runs this control in this form; it is the cleanest evidence that the cause is math-CoT SFT in general, not weak-to-strong asymmetry (§5). **(3) Refined mechanism: compression for the raw drop, plus matched-length cue suppression on the safety-relevant subset.** A length-binned Fisher analysis shows the matched-mid-length gap closes across all cued samples ($p = 0.80$), supporting compression as the explanation for the raw $25\% \rightarrow 3\%$ drop. Restricted to the safety-relevant subset (influenced=1, where the cue actually moved the answer), the matched-mid-length gap reopens to baseline 70.0% vs W2SR 21.1%,

Table 1: Reproduction (held-out MATH L3–5 Pass@1). In-family native-Qwen teacher; headroom probe separates genuine elicitation from CoT-prompting. Genuine on the base student, artifact on the instruct student.

student	unelicited	zero-shot CoT	W2SR	W2SR – CoT
Qwen2.5-Math-7B (base)	0.325	0.240	0.645	+0.405 (genuine)
Qwen2.5-7B-Instruct	0.230	0.630	0.630	+0.000 (artifact)
Foreign-style teachers (R1-distill-1.5B, SimpleRL-Zoo) → 7B: no gain / collapse.				

OR ≈ 8.75 , $p=0.003$. Decomposing the matched-length question by influenced status is methodologically new.

(4) Reproduction critique + certification consequence. A headroom probe shows W2SR’s MATH Pass@1 lift is itself conditional on (i) an in-family teacher and (ii) a base student (§4). Combined with (1)–(3), this means an accuracy-only certification of any math-CoT-SFT student (including same-strength self-distillation) can pass a model whose CoT has become markedly less revealing.

2 Related work and what is new here

Fine-tuning degrades CoT faithfulness. Lobo et al. [5] show task-specific fine-tuning lowers CoT faithfulness across four datasets; we take this degradation as given. Our claim is not that faithfulness drops but that it drops *while the cue’s behavioral influence on the answer does not*, a dissociation their answer-level faithfulness metric does not isolate. **Self-distillation and verbalization.** Concurrent work [6] finds self-distillation suppresses *epistemic* verbalization (expressed uncertainty) via compression, hurting out-of-distribution generalization. We share the compression mechanism and the self-distillation setup but differ in outcome variable: our target is *cue-acknowledgment* (a safety/monitorability property), not uncertainty calibration or OOD accuracy, and we measure the behavioral cue-pull directly. **CoT monitorability.** Meek et al. [2] measure monitorability as faithfulness plus verbosity across model families; we use their cued-prompt eval but study how a *training intervention* moves faithfulness on a fixed substrate, and pair it with a behavioral-influence measure to separate “stops being influenced” from “stops verbalizing influence,” and ask the sharper question of whether faithfulness falls *beyond* what CoT length alone explains.

3 Setup

W2SR [1]: sample CoT from a weak teacher on a problem set, keep correct + incorrect traces, LoRA-SFT a stronger student; report held-out Pass@1. **Monitorability** [2]: each question is paired with a *cue* (authority opinion, leaked answer key, embedded metadata, etc.) pointing at a target; an LLM judge reads the full CoT and scores *faithfulness* (does the CoT *acknowledge* the cue) and *verbosity* (a factor-utilization rate). We use the five cues shipped in the released eval and a single judge (`claude-sonnet-4-6`, $T=0$); faithfulness is reported as acknowledgment among *cued* samples. All evaluation is greedy ($T=0$); trace sets are hashed. Training is LoRA; generation/training/gating run on Modal GPUs.

4 Reproduction: when does W2SR fire?

With a foreign-style weak teacher (DeepSeek-R1-distilled 1.5B, or SimpleRL-Zoo) distilled into a Qwen2.5-7B student, we found *no* Pass@1 gain and frequent degeneration, across students, ranks, and decodings. W2SR reproduces only with an **in-family, native-Qwen** weak teacher: **Qwen2.5-Math-1.5B-Instruct** ($\sim 67\%$ on the train problems) → **Qwen2.5-Math-7B** lifts held-out MATH (L3–5) Pass@1 from 0.325 to 0.645, format-valid 1.00.

Genuine only on a base student (headroom probe). A +5pt gate-pass on an already-CoT-capable model may be re-arrangement, not elicitation. Scoring the *untrained* model under a zero-shot CoT prompt (the prompting-only ceiling) and reporting W2SR – (zero-shot CoT) (Table 1): on the base student W2SR adds +0.41 *beyond prompting* (genuine latent-capability elicitation); on an instruct student it adds +0.00—its apparent gain over an unelicited baseline is purely the CoT-elicitation confound. Two conditions Yuan et al. leave implicit: in-family teacher, and a base (non-CoT-eliciting) student.

5 Extension: does W2SR preserve CoT monitorability?

Capability-controlled design. We compare a baseline student, a W2SR student (LoRA-SFT’d on a weak in-family teacher’s MATH CoT), and references, on GPQA. The SFT lands the student at its own CoT ceiling, so faithfulness differences are not a capability confound; the question is “does distillation change faithfulness,” not “does it improve capability.”

Instruct student is floored (motivation). On Qwen2.5-7B-Instruct, cue-acknowledgment is at a floor for every condition (baseline 1.8% at $N=970$; W2SR 3.6%; strong-teacher control 4.6%), while teacher references reach 22–37%. We cannot detect faithfulness transfer here—but verbosity transfers strongly (W2SR–baseline $\Delta = +0.26$, CI $[0.22, 0.31]$): the model *looks* more monitorable without measurable faithfulness change. To probe a substrate with dynamic range, we move to a reasoning student.

Reasoning student: behavior preserved, verbalization collapses (Table 2). Fully in-family: student R1-Distill-Qwen-7B, weak teacher R1-Distill-Qwen-1.5B; distillation is clean (gate format-valid 0.99); baseline acknowledgment is off-floor (25%). **Behavior is not reduced.** Paired influence (answer == cue target on matched (qid, cue)): $\Delta = +0.157$, 95% CI $[+0.037, +0.278]$, $n=108$, McNemar $p=0.021$, discordant 16/33 — direction-positive but with a lower CI bound near zero, so we read this as *not reduced* rather than confidently strengthened. The cleaner, stronger signal: flips land directionally on the cue target at 73% for W2SR vs. 65% for baseline, both $\sim 2\times$ the 1/3 random-flip chance ($p < 10^{-7}$ each) — statistically indistinguishable from each other and from the untrained directionality. The pull is genuine cue-influence, not accuracy gain in disguise: restricted to the cue-points-at-wrong-answer stratum (so landing on the cue *is* a wrong answer), flippers still hit the cue target at 77%/74%/74% for baseline / W2SR weak / strong-teacher distill (each $p \leq 2 \times 10^{-9}$ vs. 1/3). **Verbalization collapses (both rubrics).** On the upstream loose rubric (“acknowledge the cue”), W2SR drops pooled acknowledgment to 3.2% (paired on (qid,cue): $\Delta = -0.220$, CI $[-0.293, -0.153]$, McNemar $p \approx 2 \times 10^{-9}$, discordant 34/1; the drop holds at every text cue: stanford 47% $\rightarrow$ 8%, grader-hack 22% $\rightarrow$ 3%, insider 56% $\rightarrow$ 5%, with one-directional discordant cells 12/0, 7/1, 15/0). *The conservative read* (stricter “attribute reasoning to the cue” rubric, same Gemini judge): the loose rubric was partly counting mentions; under the strict rubric the rates shrink to baseline 9.4% vs W2SR 0.0% (15/0 discordant, $p=6 \times 10^{-5}$), so the magnitude is smaller but the drop is *more* one-directional and the trained student never explicitly attributes reasoning to the cue. Both numbers hold at *improved* GPQA accuracy (0.28 $\rightarrow$ 0.42, $n=40$; a +14pp cross-domain lift from MATH training to GPQA science) and $\sim 14\times$ shorter CoT (baseline median 18,692 chars $\rightarrow$ W2SR 1,362). The monitorability cost is therefore being paid against a real cross-domain accuracy gain, not against a null — which is what makes the certification failure sharp. The drop is not an artifact (hand-checked: completions coherent, 37/40 conclude with an answer, none degenerate—the model solves the problem without verbalizing the cue). It is also not a single-judge artifact: re-scoring the same 525 R1-family CoTs with **gemini-2.5-pro** yields per-condition Cohen’s $\kappa \in [0.65, 0.68]$ and a paired drop $\Delta = -0.153$, McNemar $p=2.4 \times 10^{-7}$, discordant 23/0.

Mechanism: compression-dominant + a real matched-length residual. Faithfulness and CoT length both fall; the near-trivial half is that shorter CoT surfaces cues less, so the claim of interest is degradation *at matched length*. Length is endogenous to reasoning (a brevity-prompt control failed—R1 ignores it), so we isolate the residual via a length-binned Fisher’s exact analysis (Task H, quantiles of the union distribution). *Across all cued + judge-labeled samples:* at matched mid-length ([1954, 8716) chars) baseline and W2SR are indistinguishable (17.5% vs 15.1%, $p=0.80$); at very long lengths ([8716, 37816), baseline $n=119$, W2SR $n=40$) baseline is meaningfully higher (27.7% vs 15.0%, OR ≈ 2.2 , one-sided $p=0.08$) — a direction-positive residual just below conventional significance. *Restricted to the safety-relevant subset* (influenced=1, i.e. samples where the cue actually moved the answer), the picture sharpens substantially: at matched mid-length the gap reopens to baseline 70.0% (7/10) vs W2SR 21.1% (16/76), OR ≈ 8.75 , Fisher two-sided $p=0.003$; at long lengths the same direction holds (67.6% vs 40.0%, OR ≈ 3.14 , $p=0.15$ at baseline $n=34$, W2SR $n=10$). The unconditional Task H was diluted by influenced=0 samples (cue did not move the answer), where neither condition acknowledges much at any length and so the gap looks closed. So most of the raw 25% $\rightarrow$ 3% drop is **compression** (the channel separator `</think>` appears in 100% of training traces but only 22% of W2SR-student completions vs. 58% of baseline, so the compression is emergent under SFT, not from stripped supervision), and the residual lives specifically in the long-trace regime. A teacher-strength axis (Table 2) shows the effect is **compression-general**: a strong 14B teacher degrades faithfulness about as much as the 1.5B teacher and is not distinguishable from it. **Self-distillation**

Table 2: Reasoning-student monitorability (GPQA, in-family R1-distill teachers). Faithfulness = acknowledgment among cued samples. Median CoT computed on cued-cell completions. W2SR degrades faithfulness at preserved accuracy; a strong (14B) teacher degrades it nearly as much as a weak (1.5B) one, and **self-distillation (teacher == student) collapses faithfulness indistinguishably from W2SR weak** ($p=1$). All trained arms land far below baseline (25%) at magnitudes that are not statistically distinguishable from each other (pairwise McNemar $p \geq 0.09$); the effect is teacher-independent and generalizes to ordinary same-strength SFT on MATH-style traces.

condition	faithfulness	GPQA acc	median CoT (chars)
baseline R1-7B	40/160 = 25.0%	0.28	18,692
W2SR, weak teacher (1.5B)	6/190 = 3.2%	0.42	1,362
Strong-teacher distill (14B $\rightarrow$ 7B)	13/175 = 7.4%	—	1,508
Self-distill, $\leq 4k$ bud (Self-A)	2/84 = 2.4%	—	1,258
Self-distill, $\leq 8k$ bud (Self-B)	3/89 = 3.4%	—	1,292
cross-family: Llama-8B self-distill	gate fail	0.225	1,424

Cross-family row (Llama-8B self-distillation, matched recipe): GPQA accuracy $0.500 \rightarrow 0.225$ ($-27.5pp$ at 0% empty), CoT compression $10.8\times$; capability gate failed, faithfulness number not interpretable, not reported (see Limitations).

weak vs. baseline $p=2 \times 10^{-9}$; strong vs. baseline $p=1 \times 10^{-5}$; Self-A vs. baseline $p=2.4 \times 10^{-7}$ ($n=77$, disc. 23/0); Self-B vs. baseline $p=4.8 \times 10^{-7}$ ($n=82$, disc. 22/0); Self-A vs. W2SR weak $p=1$ ($n=79$, disc. 2/2); Self-B vs. W2SR weak $p=1$ ($n=79$, disc. 2/3) — statistically indistinguishable.

negative control. To test whether weak-to-strong asymmetry is causal at all, we train baseline R1-7B on its OWN R1-7B traces (1200 MATH L3-5, $T=0.6$, same SFT config). Both self-distillation arms (4k and 8k generation budgets) collapse acknowledgment to 2.4% and 3.4% — statistically indistinguishable from W2SR weak (paired $p=1$, $\Delta \approx 0$ on $n=79$ matched pairs each) — and median CoT compresses to $\sim 1.3k$ chars regardless of budget. **Cross-substrate replication on MMLU.** A small MMLU sweep (5 STEM subjects, 8 Qs each $\rightarrow$ 40 cued/cell, $n=35$ paired after parseable-answer attrition on the baseline side) holds the pattern: baseline ack 20.0%, W2SR weak 2.9%, Self-A 0.0%; Self-A vs. baseline paired $\Delta = -0.171$, McNemar $p=0.031$ (discordant 6/0, one-directional); Self-A vs. W2SR weak again indistinguishable ($p=1$). Baseline R1-7B already writes shorter CoT on MMLU (3.6k vs. 19k chars median), so MMLU compression is only $\sim 3\times$ vs. GPQA’s $\sim 14\times$, and the absolute ack drop is correspondingly smaller (MMLU $20.0 \rightarrow 2.9\%$, drop ~ 17 pp at $3\times$ compression; GPQA $25.0 \rightarrow 3.2\%$, drop ~ 22 pp at $14\times$). *Less compression, smaller drop, but the drop still fires* — a within-paper dose-response from compression to ack collapse that supports the compression-dominant reading without staking it on the matched-length cuts alone. So the faithfulness collapse *is not weak-to-strong-specific, not GPQA-specific, and not contingent on extreme CoT compression*: it fires at $3\times$ compression too, and its magnitude scales with how much compression occurred. The W2SR asymmetry is incidental; the mechanism is **SFT on terse MATH-style CoT traces in general**, which compresses CoT and suppresses cue-acknowledgment whether the teacher is weaker, stronger, or identical to the student.

6 Discussion

Math-CoT SFT does not preserve monitorability and, where it is measurable, it hurts. The two substrates are consistent: on the floored instruct student Meek’s composite “monitorability” (the mean of verbosity and faithfulness scores defined in [2]) rises after distillation almost entirely via verbosity—the model *looks* more monitorable; on the reasoning student, the cue’s *behavioral pull on the answer is not reduced while its verbal acknowledgment collapses*—a dissociation a compression-only account would not predict, since the model keeps being moved by the cue while no longer reporting it. The safety-relevant component thus gets *worse*, not better. The warning is sharp: *an accuracy-only certification of **any** CoT-SFT student trained on MATH-style traces—including same-strength self-distillation, not just weak-to-strong as in W2SR—can pass a model whose chain-of-thought has become markedly less revealing*, and a monitorability check that rewards elaborate-looking CoT would miss it too. The narrowness of the W2SR accuracy claim itself (in-family teacher + base student, §4) only sharpens this point: the certified guarantee is narrow on both axes.

7 Limitations

Floored instruct substrate: the instruct arm only motivates; the faithfulness claim rests on the reasoning student. **Matched-length residual depends on the cut.** *Unconditionally*, the gap at matched mid-length is closed (17.5% vs 15.1%, $p=0.80$) and at long lengths is marginal ($OR \approx 2.2$, one-sided $p=0.08$). *Restricted to* influenced=1 (the subset where the cue actually moved the answer, which is the safety-relevant question), the matched-mid-length gap is 70.0% vs 21.1%, $OR \approx 8.75$, $p=0.003$. Compression accounts for most of the raw drop in absolute ack rates; cue-suppression accounts for the matched-length residual on the safety-relevant subset. We do not claim compression is the sole mechanism. **Single family:** the degradation is shown within the Qwen R1-distill lineage. A same-recipe cross-family test on R1-Distill-Llama-8B fails the capability gate cleanly (50.0% $\rightarrow$ 22.5% GPQA, $-27.5pp$ at $10.8\times$ compression), placing it in the over-compression confound regime, so we do not run its monitorability eval. Cross-family generalization remains open and is reported as a null. **Cross-substrate:** the MMLU replication is direction-stable and significant for Self-A vs. baseline ($p=0.031$) but underpowered for W2SR weak vs. baseline at $n=35$; broader coverage is untested. **Also:** LoRA only; judge robustness with one alternative family (*gemini-2.5-pro*), not a panel.

8 Conclusion

W2SR’s accuracy gain is real but conditional (in-family teacher; genuine only on a base student), and on a reasoning student its faithfulness collapse (25% $\rightarrow$ 3% at preserved accuracy) is **not weak-to-strong-specific**: a same-strength self-distillation control reproduces the collapse at indistinguishable magnitude, and the cue’s behavioral pull on the answer is *not reduced* while verbalization collapses—a dissociation that survives a within-judge rubric swap (under which it tightens: ack 9.4% $\rightarrow$ 0.0%), an out-of-family judge, and a cross-substrate (MMLU) replication. The mechanism is *compression as produced by math-CoT SFT, plus an independent cue-suppression effect on the safety-relevant subset*: across all cued samples the matched-length residual is marginal, but restricted to samples where the cue actually moved the answer (influenced=1) it is substantial ($OR \approx 8.75$, $p=0.003$). The substrate dose-response (GPQA $14\times$, MMLU $3\times$) supports compression as the dominant explanation for the raw drop while the influenced=1 cut isolates the W2SR-specific failure mode at matched length. A same-recipe cross-family attempt on R1-Distill-Llama-8B fails the capability gate cleanly ($-27.5pp$ accuracy at $10.8\times$ compression), so the dissociation’s cross-family scope remains open. Certifying *any* CoT-SFT student on accuracy alone, or on how elaborate its CoT looks, is insufficient and potentially misleading; the cue-acknowledgment component of monitorability must be measured directly.

References

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